

Week 11

HDR Pathway

COS40006 Computing Technology Project B

Presented by: Ts. Selena Yap

COS40006 Computing Technology Project B

- 01** Scrum Revisited
- 02** Scrum Master
- 03** Code of Conduct
- 04** Employability
- 05** Resume & Interview Prep
- 06** Transition to Employment

- 07** CapEx
- 08** ITIL Process
- 09** Ethical AI
- 10** Project Closure & Handover
- 11** HDR Pathway
- 12** Workplace Safety

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What is

“HDR”

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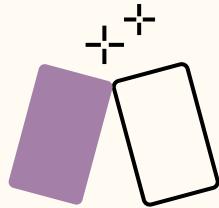
Higher Degree by Research

“HDR”

Higher Degree by Research

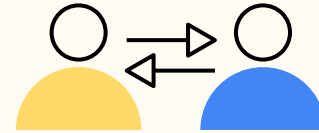
A **postgraduate** route where students pursue a Master's by Research (MRes) or a PhD instead of (or after) a taught Master's program.

HDR Programs Offered



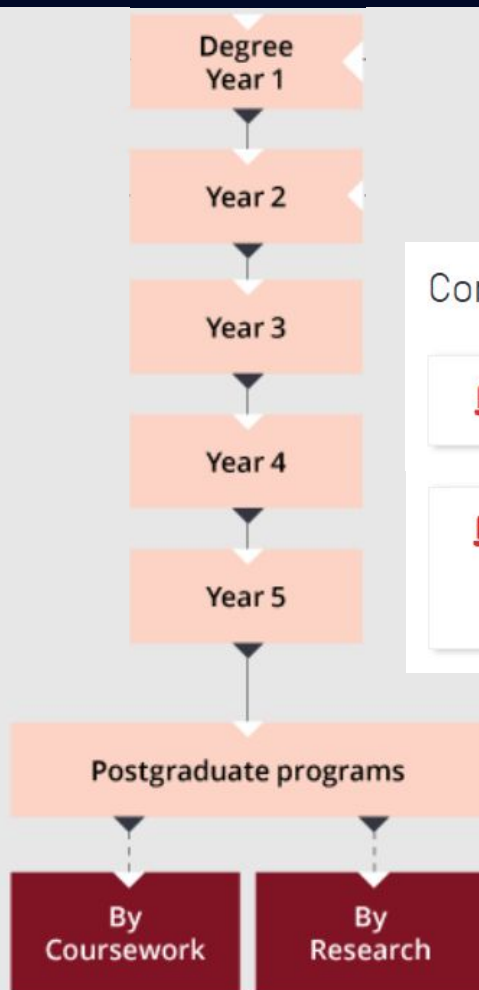
Master by Research

Requires a 3-year Bachelor's degree with $\geq 65\%$ in final year; leads to 50,000–60,000 word thesis.



PhD

Requires 4 years of tertiary study (e.g., 3-yr Bachelor + coursework Master /or/ Honours year) with high academic performance (70–100% average); involves independent research of high originality.



Computing



Master of Information Technology



Master of Information and
Communication Technologies (ICT)
(Research)



There are many paths to a PhD



bachelor's degree
+ honours degree



master's degree
with a research
component



research experience
+ relevant degree



PhD

What HDR Involves Daily

Realistic Day-to-Day Work:

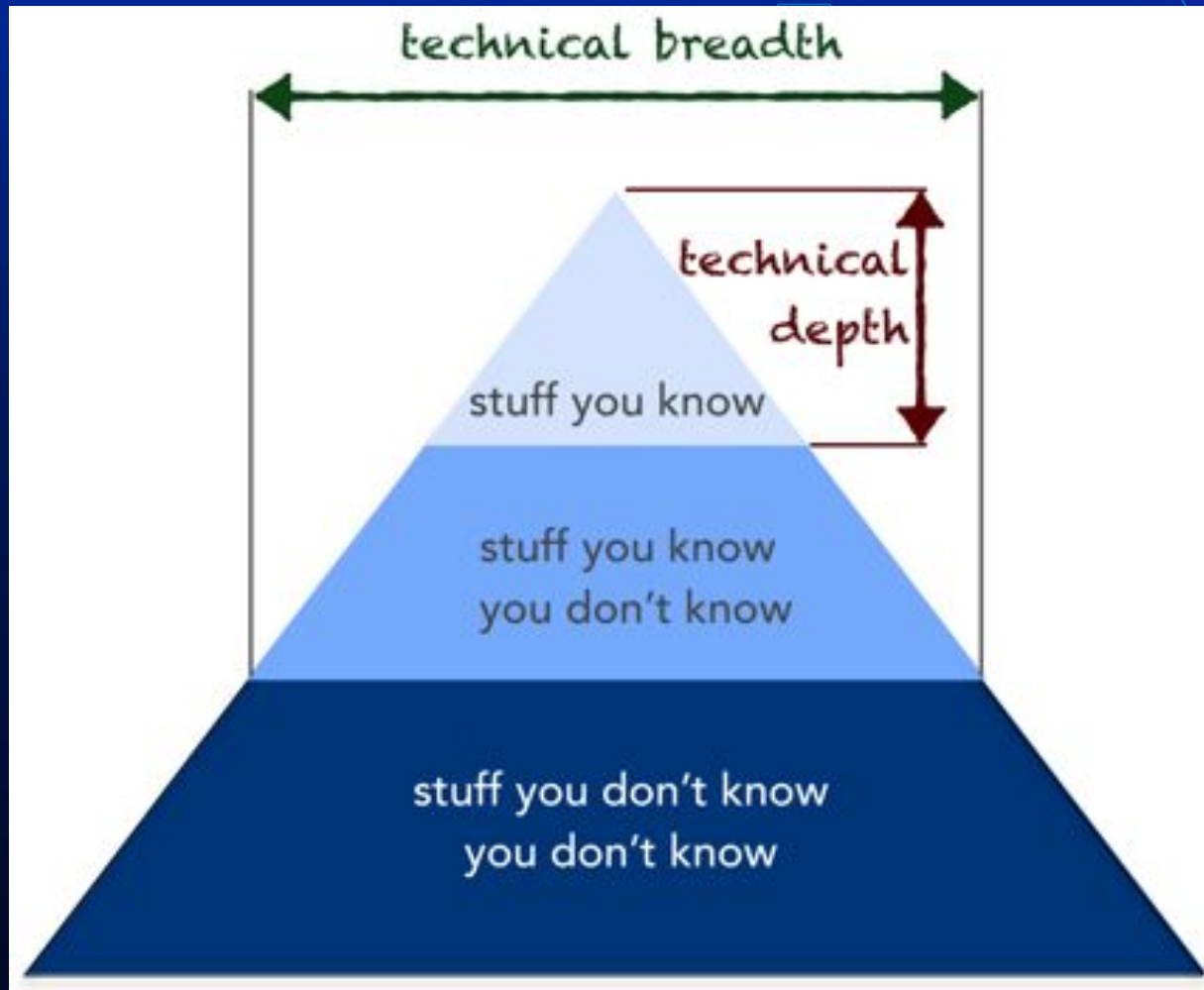
- Deep reading of research papers
- Running experiments or simulations
- Coding prototypes, models, or algorithms
- Writing technical papers.
- Presenting work to supervisors
- Data analysis, visualisation, and reporting

Tools commonly used:

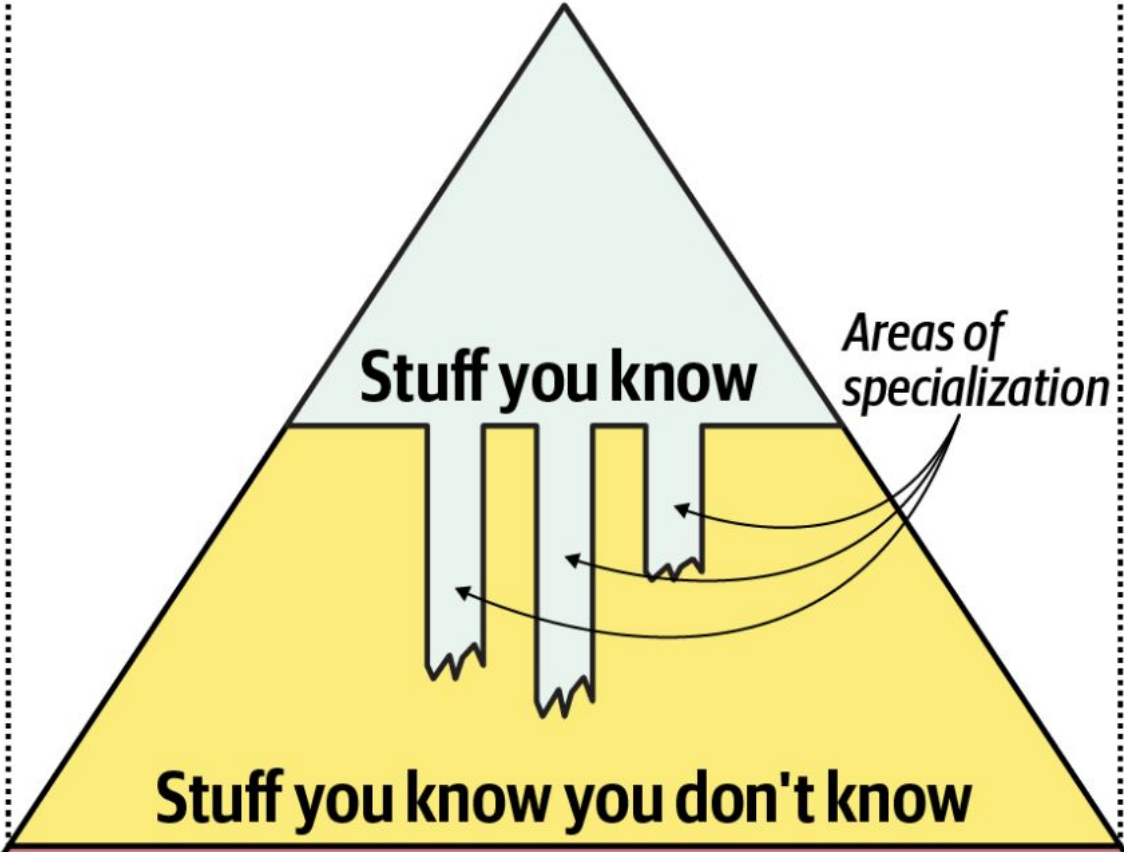
- Python, R, MATLAB, Jupyter
- IBM SPSS, SmartPLS, Statistical Software
- AWS/GCP/Azure credits for cloud experiments
- GitHub, Overleaf, Latex
- Research databases (IEEE, ACM, Springer)

Skills HDR Builds (That Industry Loves)

- **Advanced problem-solving** (solving problems with no known answers)
- **Technical depth**
e.g.: strong ML engineering, systems design, data modelling
- **Research & Analysis skills**
- **Academic writing + technical communication**
- **Time & project management** (multi-year projects)
- **Resilience** (experiments fail often)



Technical breadth





Who Should Consider HDR?

You might be a good fit if:

- You enjoy digging deeper than surface-level coursework.
- You like solving open-ended questions.
- You are curious and self-driven.
- You want to specialise in a niche (AI, cloud, security, networks, HCI).
- You don't mind writing and documentation.
- You like “slow wins” — progress comes in phases, not instantly.

🚩 Red flags if you dislike:

Lots of reading, uncertainty, long-term projects, independence.



Practical Factors to Consider

Time

HDR Master's: 1.5–2 years
PhD: 3–4+ years

Money

Scholarships *may* cover tuition + stipend, but not guaranteed.

Career Goals

- Want to be a CTO, architect, data scientist → HDR helps.
- Want to climb industry ranks quickly → Work experience matters more.



Personality Fit

- HDR = independent, self-disciplined, persistent.
- Industry = fast-paced, collaborative, structured.

How to Prepare if You're Interested in HDR

01

Identify

Identify a research topic aligned with Swinburne Sarawak's focus areas (e.g., ICT, Engineering).

02

Locate

Find a supervisor via School of Research or relevant research centre.

03

Submit

Submit application through supervisor with required documents and referee reports.

04

Follow-up

Applications are accepted year-round. Follow-up periodically for updates on your application.



Full scholarships and tuition bursaries available based on:

HDR Programs - Scholarships & Funding Opportunities

1. Academic performance
2. Research experience
3. Academic Publications
4. Alignment with university research themes



WE'RE HIRING GRADUATE RESEARCH ASSISTANT

**PROJECT ON IVF IN CATTLE
START IN NOVEMBER 2025**

**DURATION : 18 MONTH
MONTHLY ALLOWANCE
PROVIDED**

REQUIREMENTS :

- Relevant bachelor's degree with honours
- Minimum CGPA 3.0 or equivalent
- Independent and proactive
- Malaysian citizen

Apply Now

Email your CV and Degree Transcript to :
Dr. Raja El Alina Binti Raja Khalif
ariaa@umk.edu.my



WE ARE HIRING!

Graduate Research Assistant (GRA-PhD)

Project title:

Development and Evaluation of a Participatory Ergonomic Intervention in Reducing Work- Related Musculoskeletal Disorders among Oil Palm Harvesting Workers in Malaysia

Employment: Full time for 2 and half years



Job scope:

- Complete data collection and assist in documentation, data management, analysis and report writing.
- Actively communicate and participate in meetings, presentations, and other engagement with stakeholders.

Requirements:

- Master's degree (preferably related to the field of occupational safety and health).
- Must register for PhD as a full time student at University of Malaysia.
- Able to work independently and execute tasks given efficiently.

Perks:

- Opportunities for high impact publications.
- Monthly allowance (RM 2000 – 2300) based on experience and tuition fee waiver.
- Friendly and supportive work environment.

Submit your CV to:

Dr Rama Krishna
Email: rama.krishna@ummc.edu.my
Contact number: 017-3625817

Application closing date
31st March 2024



WE ARE HIRING

GRADUATE RESEARCH ASSISTANT

PROJECT TITLE:
INFUSION CRITERIA FOR MALAY MEDICINAL REQUIREMENTS IN
POSTNATAL CARE HEALING ENVIRONMENT

Allowance: RM2300/month

Duration: 2 years

DEADLINE: 30 NOVEMBER 2025

Requirements:

- Enroll as a full-time Master's and Doctorate, University of Malaysia
- CGPA 3.0 or above for a Master's
- Open to graduates from: Architecture, Interior Design, Wellness Studies, Social Science involving sports, culture, or health
- Proactive, self-motivated, and Proficient in spoken and written English

APPLY NOW

Drop your resume, transcripts
To: Nik Marul Fikri
marul.fikri@umk.edu.my

POSTGRADUATE OPPORTUNITY MASTER BY RESEARCH (FULL TIME)

Research Title : Fundamental Study on Tuning the Intrinsic Electronic State of 2D MoS₂ to Modulate the Design of Chemiresistor Hydrogen Sensor via Optical Fiber Transducer

Requirements

- Must enroll as full time GRA Master Sc. in Engineering at UNIMY
- CGPA 2.8 and above – Bachelor Degree in Electrical/Electronic/Communication/Material Engineering with good English command of spoken and written
- Ability to work independently, committed and good attitude

Benefits

- Monthly allowance of RM 2000 for 2 years
- Immediate vacancy
- Research is fully funded under FRGS scheme (MOHE)

Contact

Dr. Azlinah Md Lazam
marazlinah@unimy.edu.my
School of Engineering & Frontier Technology





GRADUATE RESEARCH ASSISTANT VACANCY FULL-TIME MASTER BY RESEARCH

Project Title:
Comparative Study of Soil Properties in the Converted Sabal Bamboo Forest and the Adjacent Secondary Forest

APPLY NOW!

REQUIREMENTS:

- Malaysian citizen.
- Bachelor's degree in Science/ Plant Science/ Forestry/ Environmental Science/ Ecology/ Soil Science/ Agriculture/ Bioindustry Science or any relevant field with a minimum CGPA of 3.00.
- Strong interest and passionate in research.
- Good scientific writing skills, able to work independently and in a team.
- Proficient in English (written & spoken).
- Able to provide full commitment.

OTHER DETAILS:

- Monthly allowance will be provided.
- Project duration: Two (2) Years
- Enrolment: March 2025.

Send your CV/ Resume to:
Dr Mugunthan Perumal
pmugunthan@unimas.my

In collaboration with

PUSAKA

Shortlisted candidates will be contacted for an interview

Application deadline: 28th February 2025

For more information: +6082-58-2940 www.fst.unimas.my

Academic Community & Support

- Swinburne Sarawak hosted an inaugural HDR conference, featuring 60+ presentations and 120 enrolled HDR students.
- Active **Postgraduate Research Society (PGRS)** organizes social, career, and mental health events for research students.
- **School of Research** emphasizes high-impact, engaged research aligned with Sarawak's development strategy and Swinburne's Vision 2025.



Benefits of HDR Pathway

- Transition from implementing systems to creating new knowledge via thesis-driven inquiry.
- Gain deep research skills: literature review, methodology, analysis, academic writing.
- Access to funding, supervision, professional development + community of peers
- Exposure to industry-relevant innovation and high-level problem-solving.

Immediate Next Steps for Students

STEP
01

Explore Master by Research or PhD if interested in R&D, academic, or deep-tech careers.



STEP
02

Prepare strong academic transcripts and consider research aligning with ICT supervisors.



STEP
03

Reach out to the School of Research or HDR supervisors early for guidance.



STEP
04

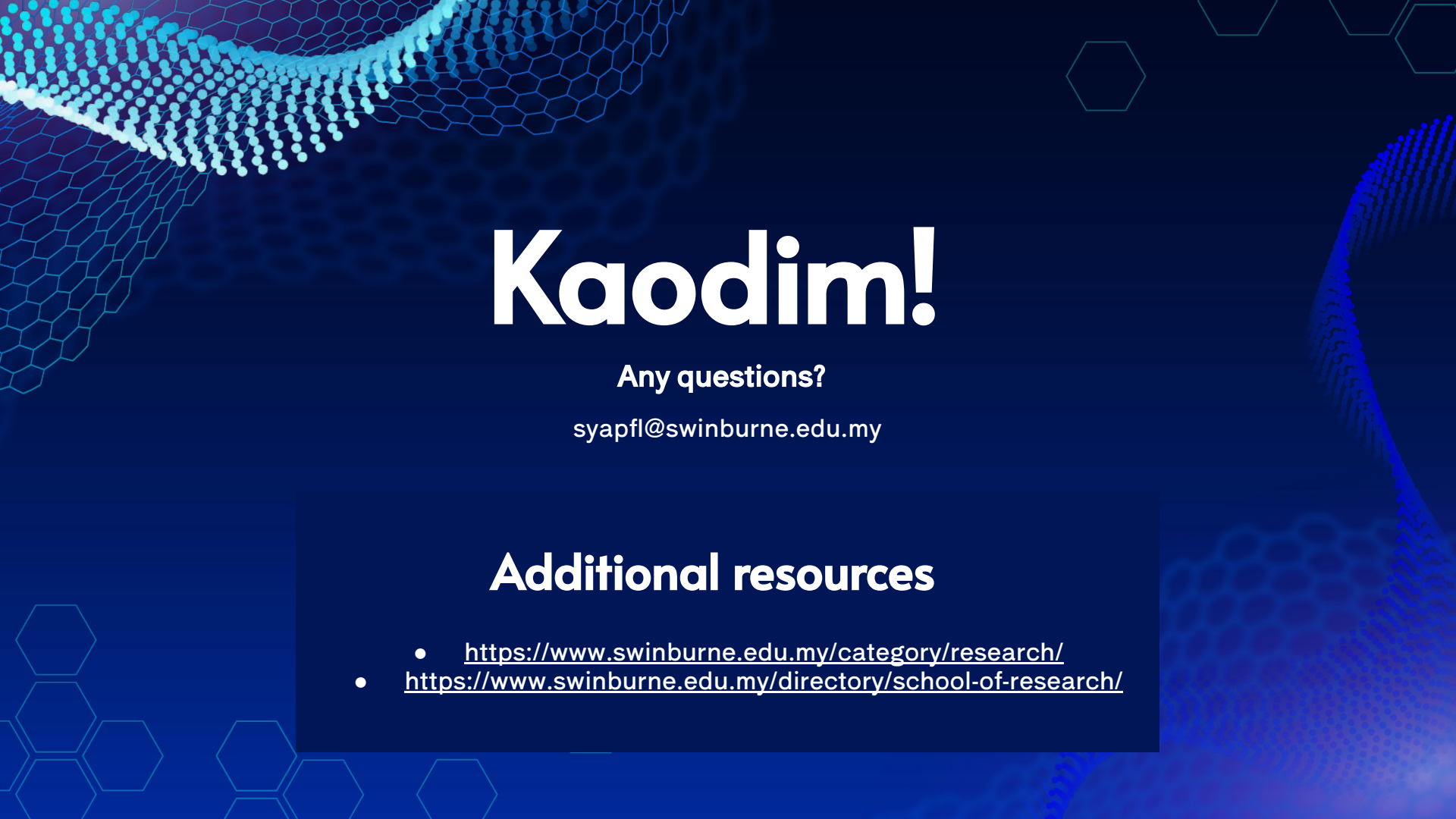
Monitor opportunities like Brooke Heritage Trust Fellowship, and engage with PGRS for community and support.





Learn More:

<https://www.swinburne.edu.my/research-degrees/>



Kaodim!

Any questions?

syapfl@swinburne.edu.my

Additional resources

- <https://www.swinburne.edu.my/category/research/>
- <https://www.swinburne.edu.my/directory/school-of-research/>